

TAIWO BAMGBOYE

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RESEARCH INTERESTS

Power System Operations, Energy Systems, Power System Optimization, Control systems, Power grid Monitoring and Control, Infrastructure Resilience, Machine Learning Applications in Power and Energy Systems.

EDUCATION

University of Ibadan

Ibadan, Nigeria

BSc. Electrical & Electronic Engineering

2019 - 2025

Grade: First Class Honors (Summa cum laude)

Thesis: Hybrid optimization of distribution networks: Feeder reconfiguration and distributed generation siting using metaheuristic algorithms for power loss reduction and voltage stability.

SKILLS AND COMPETENCES

- Programming Languages: Python, Java, C/C++, Vhdl, Fortran, Mathematica
- ML & Data Science: Deep Learning, Reinforcement Learning (DRL), Neural Networks, TensorFlow
- Layout Design Software: Proteus, KiCad
- Circuit/System Software: Cadence-Affirma/Spectre, Simulink
- Numerical Solvers: Coventorware, Femlab, CFD-ACE, Fluent, Adams/AView
- Mathematical Packages: Matlab, Mathematica, Maple CAD Software: SolidWorks

RESEARCH EXPERIENCES

Power, Energy, Machines and Drives (PEMD) Research Group

June 2024- April 2025

Undergraduate Research Assistant

University of Ibadan, Ibadan

Principal Investigators: Prof A.S.O. Ogunjuyigbe & Prof T. R. Ayodele

Project title: Hybrid optimization of distribution networks: Feeder reconfiguration and distributed generation siting using metaheuristic algorithms for power loss reduction and voltage stability.

Summary: Despite extensive research in power distribution systems, Nigerian networks continue to exhibit persistent challenges such as excessive power losses and voltage instability due to limited adoption of advanced optimization techniques. This study employs a Genetic Algorithm-based multi-objective optimization framework for simultaneous network reconfiguration and DG placement, yielding substantial improvements in voltage profiles and loss minimization across 33kV distribution networks and IEEE benchmark systems.

Duties: Developed and implemented a Genetic Algorithm for optimal DG placement and network reconfiguration. Validated results on IEEE and 33kV Nigerian feeders; analyzed improvements in power loss and voltage stability using MATLAB.

MANUSCRIPTS IN PREPARATION & IN REVIEW

Bamgboye T., Ogunjuyigbe A.S.O., Ayodele T. R. Optimal sizing and placement of distributed generation in radial distribution networks using genetic algorithms with application to Nigerian feeders. *In prep for Applied Energy*.

Eniola A. Oladele, Israel T. Olanipekun, **Taiwo Bamgboye**, Akinsola O. Emilayokan, Ibrahim Oreagba. Knowledge, Attitude and Practice of Nurses in South West Nigeria Towards Adverse Drug Reaction Reporting: A Multicenter Cross-Sectional Study. **West African Journal of Pharmacy (In review)**

WORK EXPERIENCE

Cydene Energy Services - Graduate Engineer

June 2025 - present

- Design and Implementation of projects involving petrol-to-CNG conversion, electricity metering, and gas cylinder manufacturing.

Rajtronics Technologies — Embedded Systems Intern

Jan - June 2024

- Designed and programmed IoT-based solutions including home automation systems, Smart GPS Waste Bins, and Agricultural Irrigation Systems (AgricBot) integrating sensors, microcontrollers, and communication modules
- Implemented circuit designs and PCB layouts for multiple IoT automation and optimization projects.

Digitouch— Embedded Systems Intern

May - Aug. 2023

- Programmed Atmega and PIC microcontrollers for IoT applications including smart waste management and irrigation control systems
- Co-developed energy solutions including digital displays and programmable moving message boards

Rashad Link Limited— Engineering Intern

Nov. 2021 -2022

- Performed maintenance and repair of industrial generators (Mikano series); executed coil rewinding for motors and transformers
- Conducted electrical installations and troubleshooting in commercial and industrial facilities maintaining compliance with safety standards

Robotics and Artificial Intelligence, Nigeria — Project Intern

Feb -April 2020

- Designed voice-recognition interface for automated restaurant food ordering system using natural language processing

TEACHING EXPERIENCE

Academic Tutor/Mentor – Dept. of EE Engineering, University of Ibadan

July 2024 - May 2025

- Provided one-on-one academic coaching and mentorship to undergraduate students in Math and Power Systems courses.

Academic Mentor – Peer Assisted Study Sessions (PASS), University of Ibadan

2023 - 2024

- Led peer-assisted tutorials and academic mentoring for three cohorts of 100-level students resulting in over 75% improvement in student performance.

Volunteer Tutor – Heroes and Heroines Initiative (University of Ibadan Chapter)

2022 - 2024

- Taught Mathematics and Physics to over 1,000 students in underserved secondary schools across Ibadan.

LEADERSHIP & VOLUNTEERING EXPERIENCE

Society of Electrical and Electronic Engineering SRC,— Speaker

July 2024 - May 2025

- Recommended constitutional reforms introducing new governance roles; organized legislative educational visits to Oyo State and Lagos State Houses of Assembly for 50+ engineering students; launched gender inclusivity initiatives within the engineering community.

Students' Union University Of Ibadan — Electoral Commissioner

July 2024 - May 2025

- Coordinated free and transparent student elections; collaborated with logistics and screening ; organized voter education campaigns increasing procedural awareness.

Society of Electrical and Electronic Engineering(SEEES) UI, — General Secretary

Nov.- July 2023

- Managed organizational communications and social media platforms, increasing engagement by over 40%; drafted press releases and represented organizations at campus journalism events.

Google Developer Groups on Campus(GDGC), Frontend Team Lead

Jan - Aug. 2023

- Led development of user-centric frontend solutions; mentored team members in React.js and nodejs; organized coding workshops and represented GDGC at campus technology events.

PROFESSIONAL ORGANIZATIONS

Graduate Member, Nigerian Society of Engineers

2025 - present

Student Member, Institute of Electrical Electronic Engineering

2019 - present

Member, Junior Chamber International Nigeria, University of Ibadan(JCIN)

2022 - present

HONOURS AND AWARDS

Dean's Award for First class Graduates

April 2025

Overall best student in Power Electronics, Current Trends Technologies

April 2025

Department of E & Electronic Engineering Award for Outstanding Performance(B.Sc Hons)

April 2025

NNPC/SEPLAT JV National Undergraduate Scholarship

June 2021

Dean's Honors list, Faculty of Technology

2018 - 2025

FELLOWSHIPS

Fellow, Campus Lab Employability Accelerator Programme

May 2021